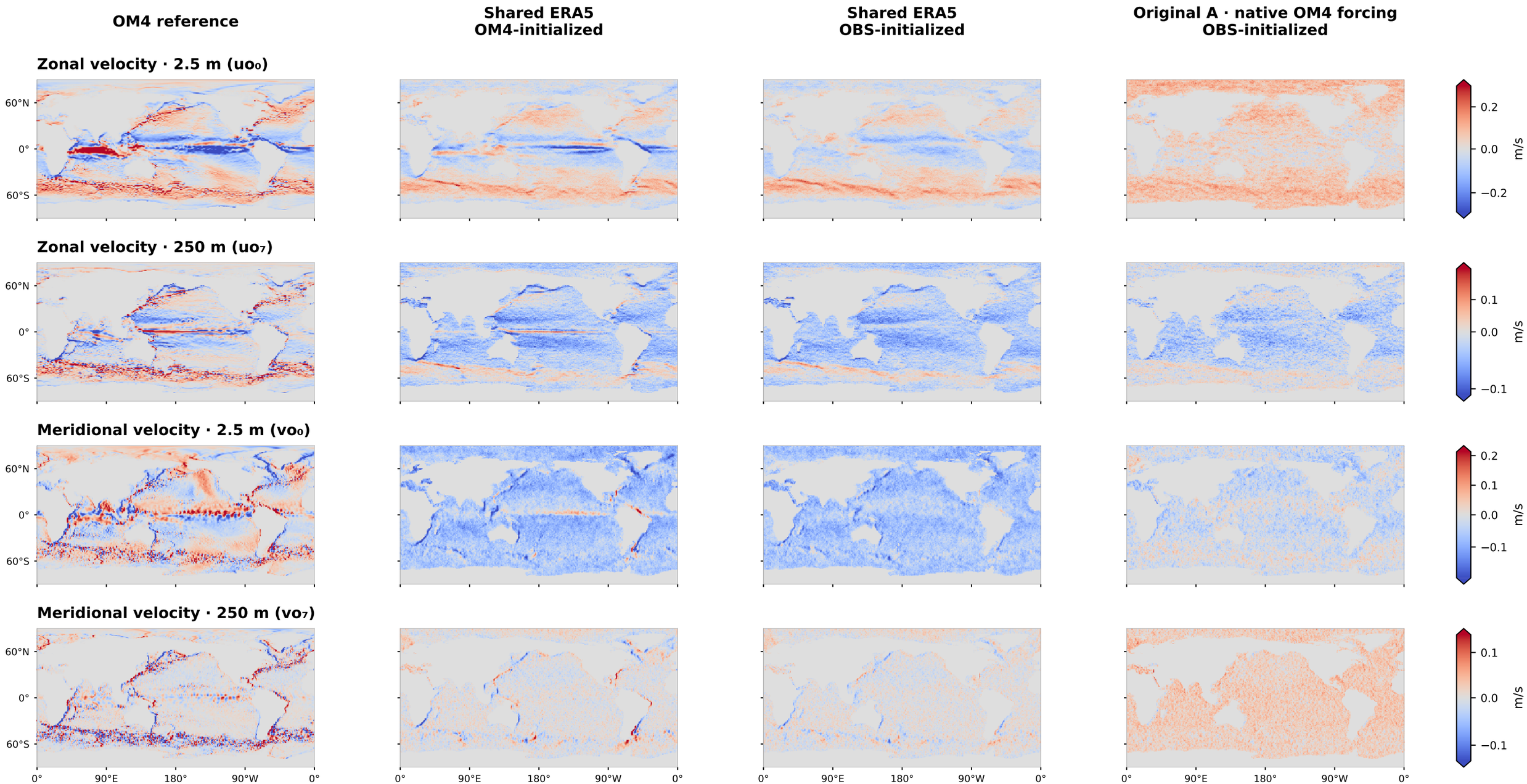
Ocean velocity | Initialization · 8-member mean

Date-matched five-day fields: 27 Dec 2014–1 Jan 2015 (exclusive end) · EMA weights · eight-member diffusion mean



Shared ERA5: 4k mixed updates + 1k OBS cooldown. Original A: 2k mixed updates + 1k OBS cooldown; approximate comparison, not a matched forcing ablation.

“Native OM4 forcing” describes Original A’s OM4 training tasks. All OBS-initialized examples use ERA5 through their learned adapters.

OBS velocity outputs are full-77 diagnostic decodes, not directly supervised OBS currents. OM4 reference is simulation context, not observational truth.

Each row uses OM4 reference 2nd–98th percentile bounds for all columns; tails clipped. Gray = dry cells. Eight decoder draws per fixed latent state; 32 Heun steps each.